Data Science and Artificial Intelligence

Machine Learning

Multi layer Perceptron

Lecture No. 4

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Recap of Previous Lecture

- Topic
- Topic
- Topic
- Topic
- Topic

FFNN

Topics to be Covered

- Topic — Back Propagation
- Topic — Numericals & Questions
- Topic
- Topic
- Topic

REVISION

FFNN →

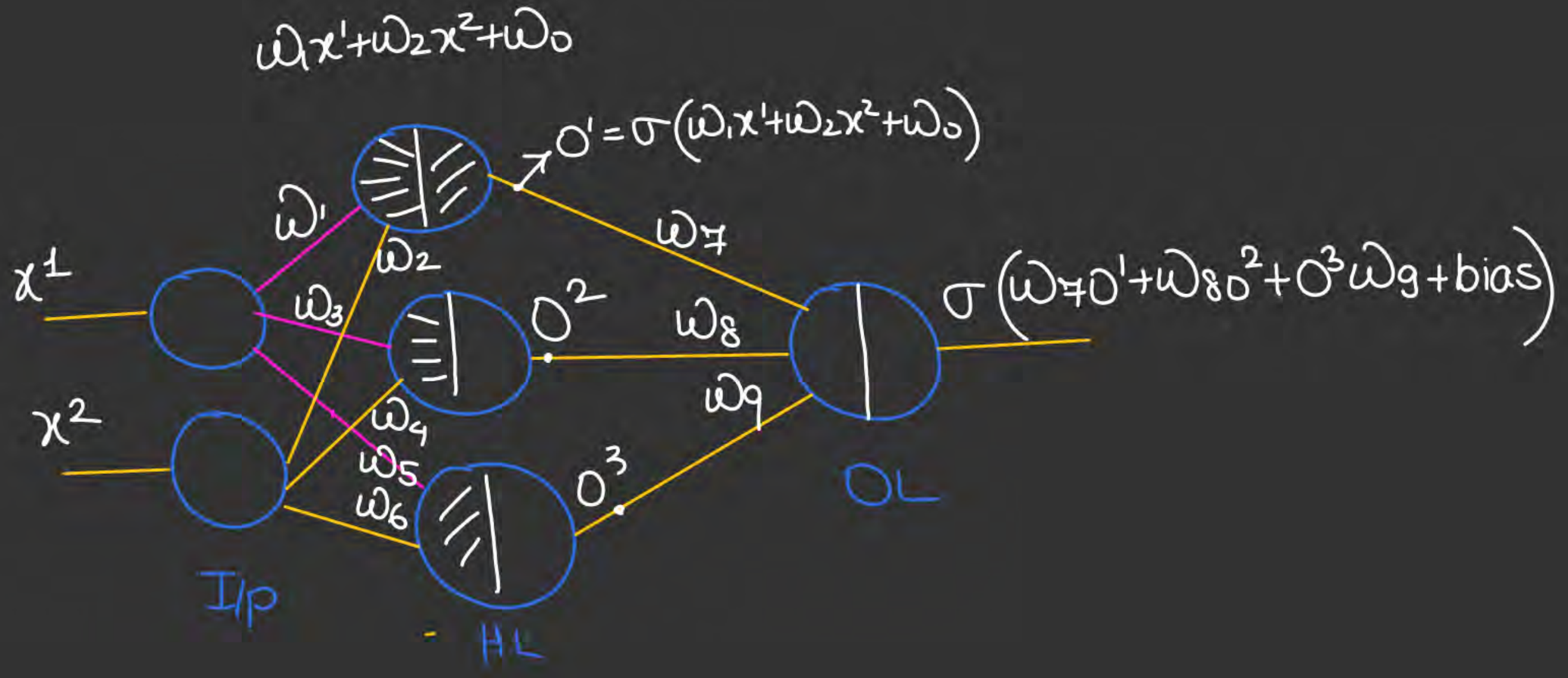
1. Can be used for R & C
2. No of weights
3. all neurons of a layer are connected to " " " next layer ⇒ Yes
4. If no limitation on No of neurons then overfit

REVISION

FFNN

(5) Only 3 layer system I/P, Hidden, O/P are enough to learn any NL fxn

(6) Softmax → If I have numbers (5, 9, 10, 12 - - -)
↳ Probab


$\omega_1 x^1 + \omega_2 x^2 + \omega_0$
$O^1 = \sigma(\omega_1 x^1 + \omega_2 x^2 + \omega_0)$
x^1
x^2
ω_1
ω_2
ω_3
ω_4
ω_5
ω_6
ω_7
ω_8
ω_9
O^2
O^3
$\sigma(\omega_7 O^1 + \omega_8 O^2 + O^3 \omega_9 + bias)$
I/P
HL
OL

Perceptron learning

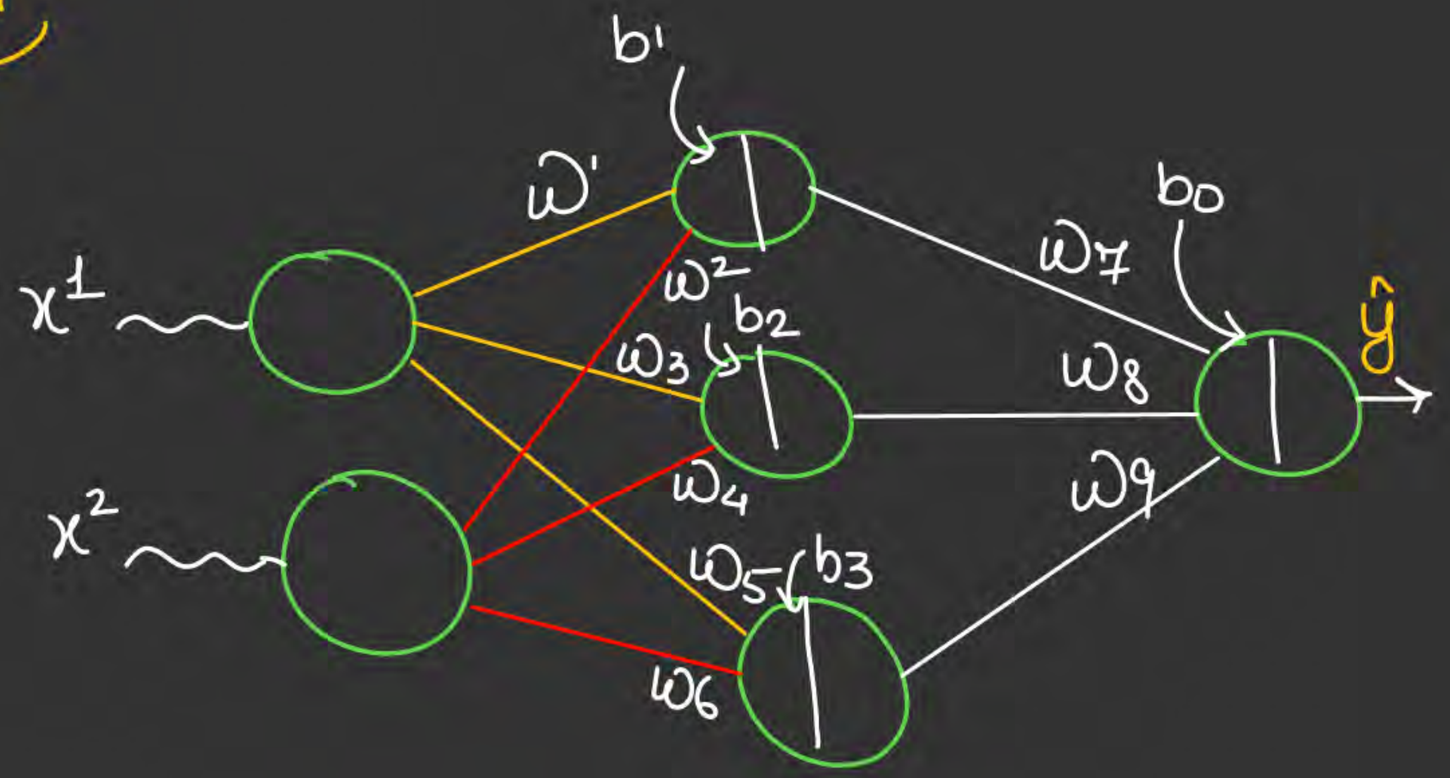
Steps in Feedforward Neural Networks Work

- Initialization: Initialize weights ww and biases bb to small random values.
- Forward Propagation: Calculate the output of the network using the current weights and biases.
- Loss Calculation: Compute the loss (error) using a loss function such as mean squared error (MSE) for regression or cross-entropy for classification.
- Backpropagation: Calculate the gradient of the loss with respect to each weight using the chain rule.
- Update the weights and biases to reduce the loss using an optimization algorithm like gradient descent.
- Iteration: Repeat forward propagation, loss calculation, and backpropagation for multiple epochs until the network converges or a stopping criterion is met.

Backpropagation

$$\mathcal{L} = \frac{1}{2}\left(y_i - \hat{y}_i\right)^2$$

$$W = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ \vdots \\ w_9 \\ b_0 \\ b_1 \\ \vdots \\ b_3 \end{bmatrix} \cdot 13 \text{ Parameters}$$



$$w^{new} = \left(w^{old} - \eta \frac{\partial L}{\partial w^{old}}\right)$$

So one input is applied, we find $\hat{y}$ and then find error and then find $\frac{\partial L}{\partial w^{old}}$ for all parameters

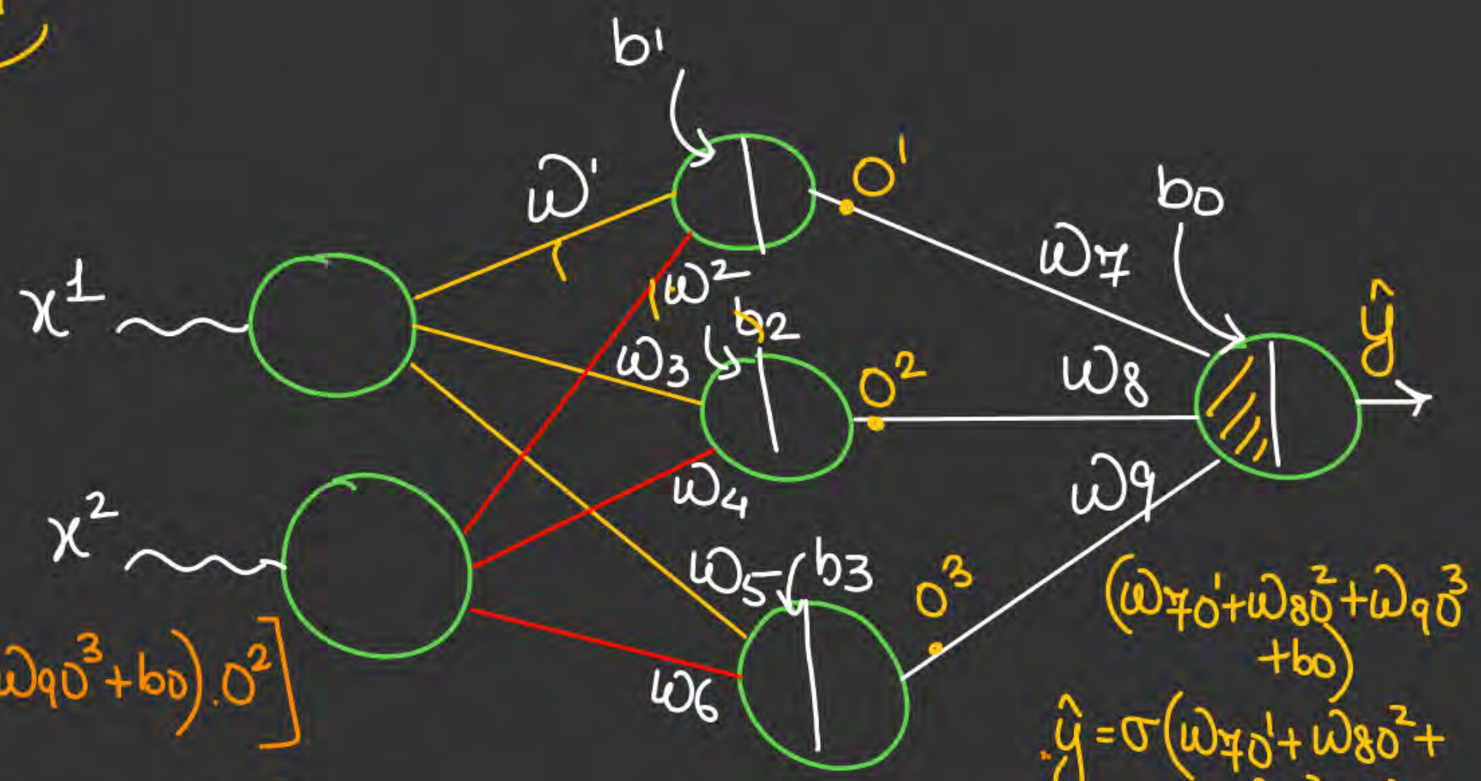
then we find $w^{new} = \left(w^{old} - \eta \frac{\partial L}{\partial w^{old}}\right)$

So after updating all w's again, next training data point is given as input.

This cycle will Run "N" times.

Backpropagation

$$L = \frac{1}{2}(y_i - \hat{y}_i)^2$$



$$(w_7 o^1 + w_8 o^2 + w_9 o^3 + b_0)$$

$$\hat{y} = \sigma(w_7 o^1 + w_8 o^2 + w_9 o^3 + b_0)$$

$$\frac{\partial L}{\partial w_8} = \frac{\partial L}{\partial \hat{y}_i} \cdot \frac{\partial \hat{y}_i}{\partial w_8}$$

$$= \left[-(y_i - \hat{y}_i) \cdot \sigma'(w_7 o^1 + w_8 o^2 + w_9 o^3 + b_0) \cdot o^2\right]$$

$$\frac{\partial L}{\partial w_7} = \frac{\partial L}{\partial \hat{y}_i} \cdot \frac{\partial \hat{y}_i}{\partial w_7} \Rightarrow -\left[(y_i - \hat{y}_i)\sigma'(----)o^1\right]$$

$$\frac{\partial L}{\partial w_9} \Rightarrow \frac{\partial L}{\partial \hat{y}_i} \cdot \partial \hat{y}_i / \partial w_9 \Rightarrow \left[-(y_i - \hat{y}_i)\sigma'(---)o^3\right]$$

Perceptron learning

What is Back Propagation

- Weights updating formula :

$$\left(w^{new} = w^{old} - \eta \frac{\partial L}{\partial w^{old}} \right)$$

What is Back Propagation

- We have a loss function :

- We use Optimizer to reduce the error (Gradient Descent)

$\eta = .5$

$y = .01$

w_1 .15
w_2 .2
w_3 .25
w_4 .30

$w_5 = .4$
$w_6 = .45$

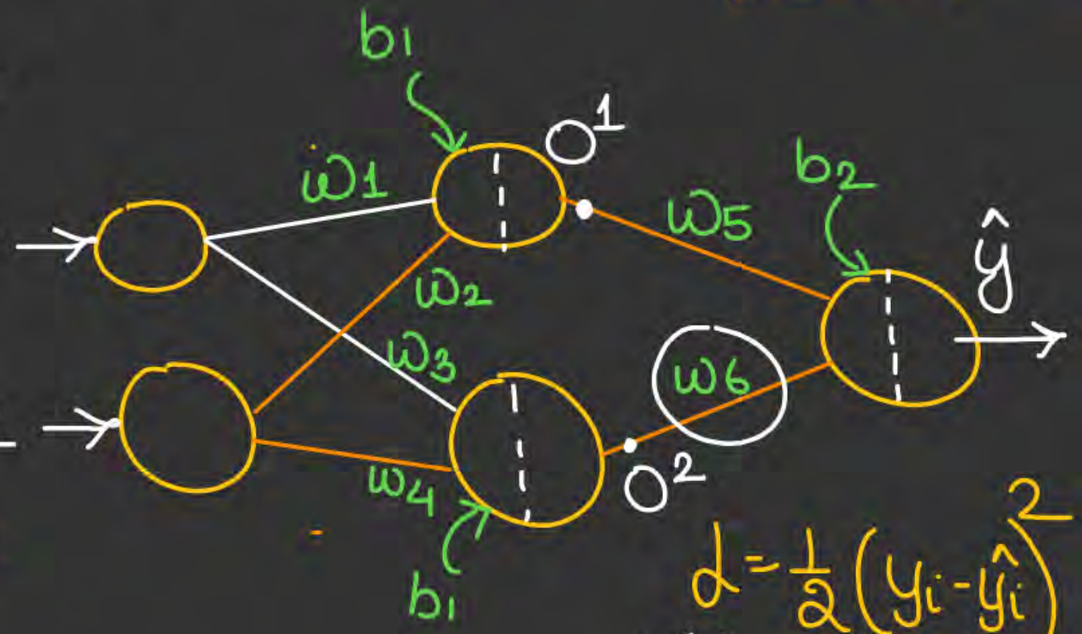
$b_1 = .35$
$b_2 = .6$

$\sigma(x) = \max(0, x)$

$$O^1 = \sigma(\underbrace{w_1 x^1 + w_2 x^2 + b_1}_{.3775}) = .3775$$

$$O^2 = \sigma(\underbrace{w_3 x^1 + w_4 x^2 + b_1}_{.3925}) = .3925$$

$$\hat{y} = \sigma(w_5 O^1 + w_6 O^2 + b_2) = .927625$$

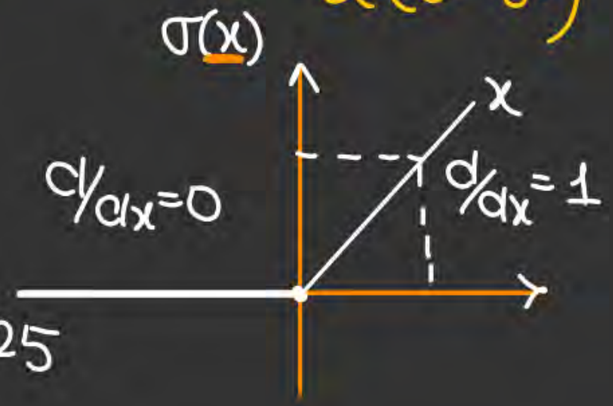


$$d = \frac{1}{2}(y_i - \hat{y}_i)^2$$

$$w_6 \Rightarrow \frac{\partial L}{\partial w_6} \Rightarrow \frac{\partial L}{\partial \hat{y}_i}\frac{\partial \hat{y}_i}{\partial w_6} \Rightarrow -(y_i - \hat{y}_i)\, 1\, O^2$$

$$\Rightarrow -(.01 - .927625) \times .3925$$

$$\Rightarrow .3601$$



Network Structure

- Input layer: 2 neurons
- Hidden layer: 2 neurons with sigmoid activation
- Output layer: 1 neuron with sigmoid activation

Given Data

- Input: $\mathbf{x} = [0.05, 0.10]$
- Target output: $y = 0.01$
- Learning rate: $\eta = 0.5$

Initial Weights

- Input to hidden layer weights:

$$w_1 = 0.15, \quad w_2 = 0.20, \quad w_3 = 0.25, \quad w_4 = 0.30$$

- Hidden to output layer weights:

$$w_5 = 0.40, \quad w_6 = 0.45$$

Biases

- Hidden layer biases: $b_1 = 0.35$
- Output layer bias: $b_2 = 0.60$

$\eta = .5$

$y = .01$

w_1 .15
w_2 .2
w_3 .25
w_4 .30

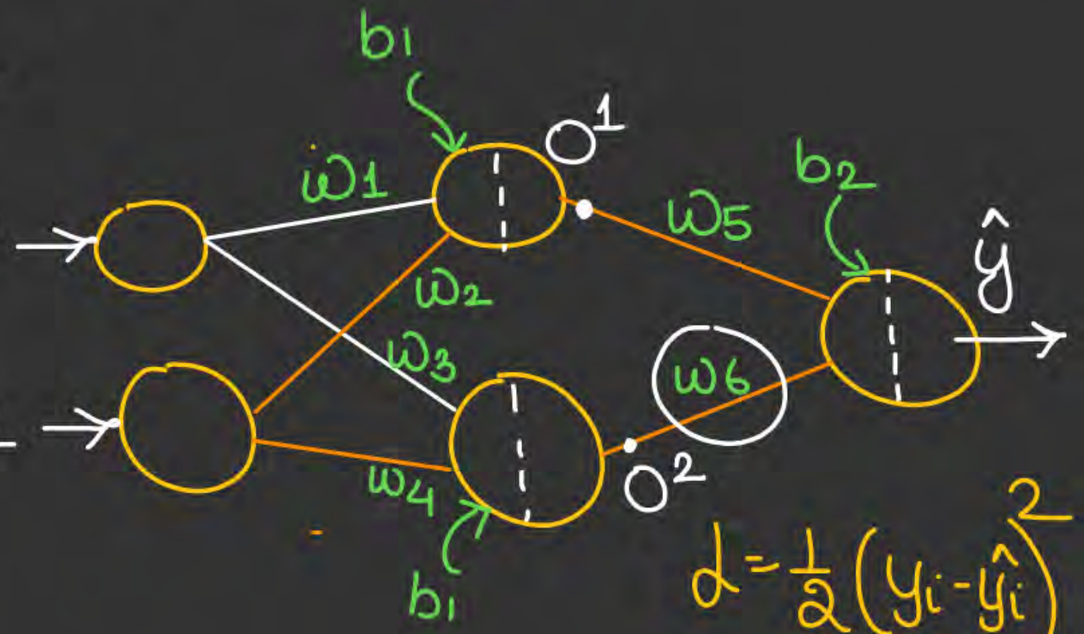
$w_5 = .4$
$w_6 = .45$

$b_1 = .35$
$b_2 = .6$

$$O^1 = \sigma\Big(\underbrace{w_1 x^1 + w_2 x^2 + b_1}_{.3775}\Big) = .59326$$

$$O^2 = \sigma\Big(\underbrace{w_3 x^1 + w_4 x^2 + b_1}_{.3925}\Big) = .596$$

$$\hat{y} = \sigma\Big(\underbrace{w_5 O^1 + w_6 O^2 + b_2}_{1.1055}\Big) = .7512$$



$$d = \frac{1}{2}(y_i - \hat{y}_i)^2$$

$$w_6 \Rightarrow \frac{\partial L}{\partial w_6} \Rightarrow \frac{\partial L}{\partial \hat{y}_i}\frac{\partial \hat{y}_i}{\partial w_6} \Rightarrow -(y_i - \hat{y}_i)\,\sigma'\big(\underbrace{w_5 O^1 + w_6 O^2 + b_2}\big)\,O^2$$

$$\Rightarrow -(.01 - .7512)\,.596\,\frac{e^{-1.1055}}{(1+e^{-1.1055})^2} = .0825$$

So $w^{6\,new} = w^{6\,old} - \eta \times \frac{\partial L}{\partial w_6}$

$= .45 - .5 \times .0825$

$= .40875$

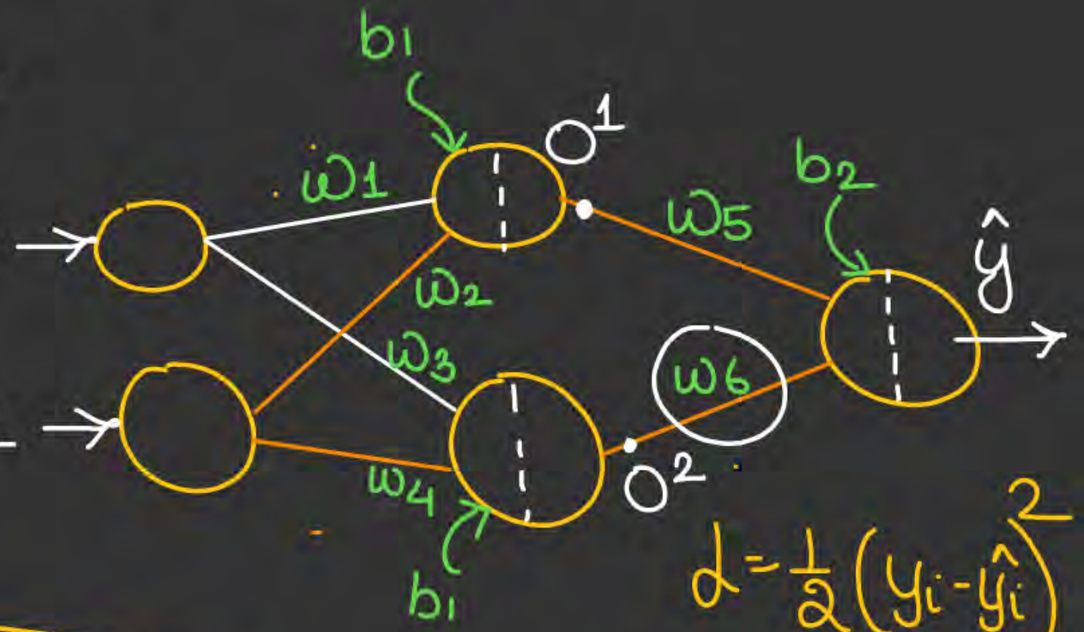
$\eta = .5$

$y = .01$

w_1 .15
w_2 .2
w_3 .25
w_4 .30

$w_5 = .4$
$w_6 = .45$

$b_1 = .35$
$b_2 = .6$



$$O^1 = \sigma\Big(\underbrace{w_1 x^1 + w_2 x^2 + b_1}_{.3775}\Big) = .59326$$

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$$\hat{y} = \sigma\Big(\underbrace{w_5 O^1 + w_6 O^2 + b_2}_{1.1055}\Big) = .7512$$

$$d = \frac{1}{2}\left(y_i - \hat{y}_i\right)^2$$

$$\frac{\partial L}{\partial w^1} = \frac{\partial L}{\partial \hat{y}_i} \cdot \frac{\partial \hat{y}_i}{\partial O^1} \cdot \frac{\partial O^1}{\partial w^1} \Rightarrow -\underbrace{(y_i - \hat{y}_i)}_{-(.01-.7512)} \; \underbrace{\sigma'(w_5 O^1 + w_6 O^2 + b_2)}_{\frac{e^{-1.1055}}{(1+e^{-1.1055})^2}} \; \underbrace{w^5}_{*\,.4} \; \underbrace{\sigma'(w_1 x^1 + w_2 x^2 + b_1)}_{*\,\frac{e^{-.59326}}{(1+e^{-.59326})^2}} \; \underbrace{x^1}_{\times .05}$$

$$= 6.3495 \times 10^{-4}$$

What is Back Propagation

- Lets learn the back propagation…

$$\left(\sigma(w^T x) = \frac{1}{1+e^{-w^T x}}\right) \rightarrow \frac{e^{-w^T x}}{(1+e^{-w^T x})^2}$$

$$f(x) = \frac{1}{1+e^{-x}} = \frac{e^{-x}}{(1+e^{-x})^2}$$

$$f(w^T x) = \frac{1}{1+e^{-w^T x}} = \left(\frac{e^{-w^T x}}{(1+e^{-w^T x})^2} \cdot \frac{\partial w^T x}{\partial \cdots}\right)$$

$$f(\sin(x))$$

$$\downarrow$$

$$\frac{d}{dx} f(\sin(x))$$

$$\Downarrow$$

$$f'(\sin(x))(-\cos(x))$$

$$f(x) = \sin(6y + 8z + 10w + 8)$$

$$\frac{\partial f(x)}{\partial z} \Rightarrow \frac{\partial \sin(\overbrace{6y + 8z + 10w + 8}^{\alpha})}{\partial z}$$

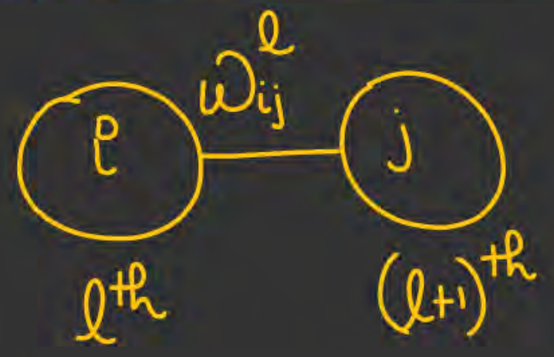
$$\alpha = 6y + 8z + 10w + 8$$

$$\Rightarrow \frac{\partial \sin\alpha}{\partial z}$$

$$\Rightarrow \frac{\partial \sin\alpha}{\partial \alpha} \cdot \frac{\partial \alpha}{\partial z}$$

$$\Rightarrow +\cos(\alpha)\, 8$$

Why it is called back propagation



Why it is called back propagation

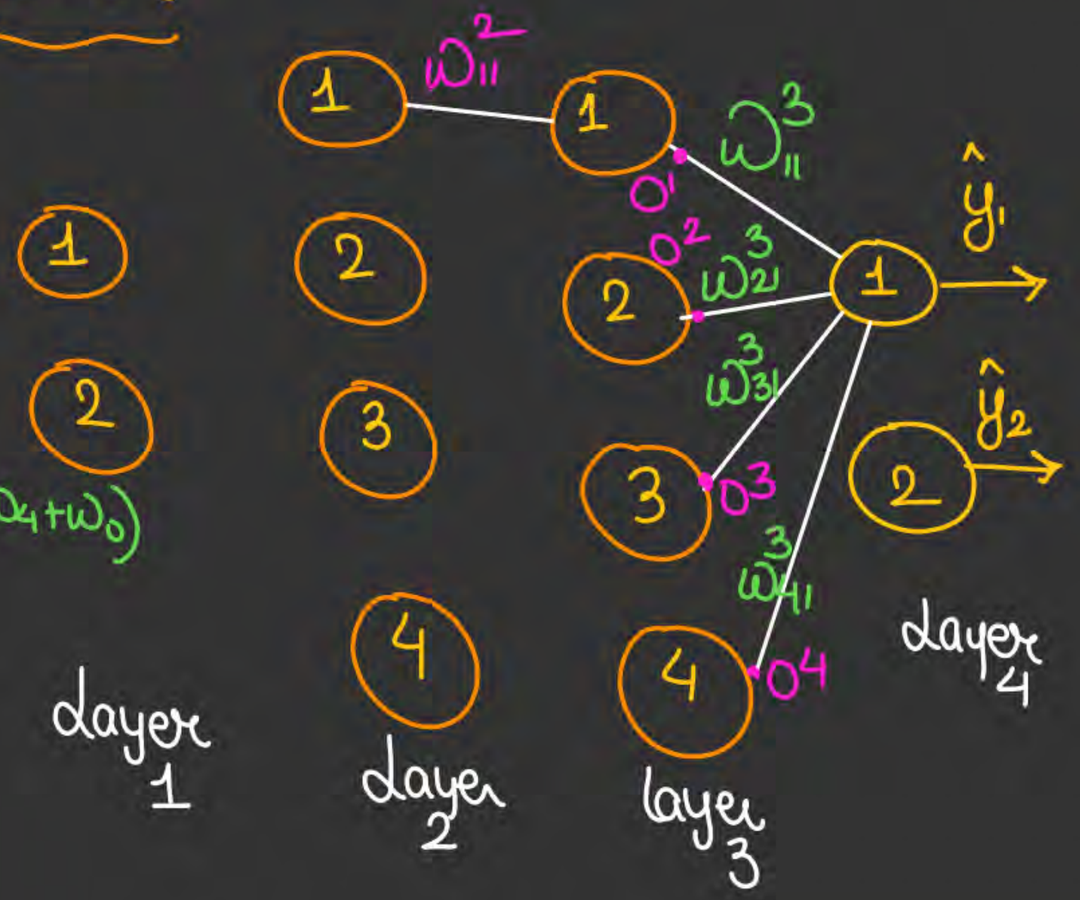
$$\mathcal{L} = \frac{1}{2}\begin{bmatrix}(y_1-\hat{y}_1)^2 \\ (y_2-\hat{y}_2)^2\end{bmatrix}$$

$$\frac{\partial L}{\partial w_{11}^3} \Rightarrow \frac{\partial L}{\partial \hat{y}_1}\cdot\frac{\partial \hat{y}_1}{\partial w_{11}^3}$$

$$\Rightarrow \left[-(y_1-\hat{y}_1)\,\sigma'(\quad).O^1\right]$$

$$(w_{11}^3 O_1 + w_{21}^3 O_2 + w_{31}^3 O_3 + w_{41}^3 O_4 + w_0)$$

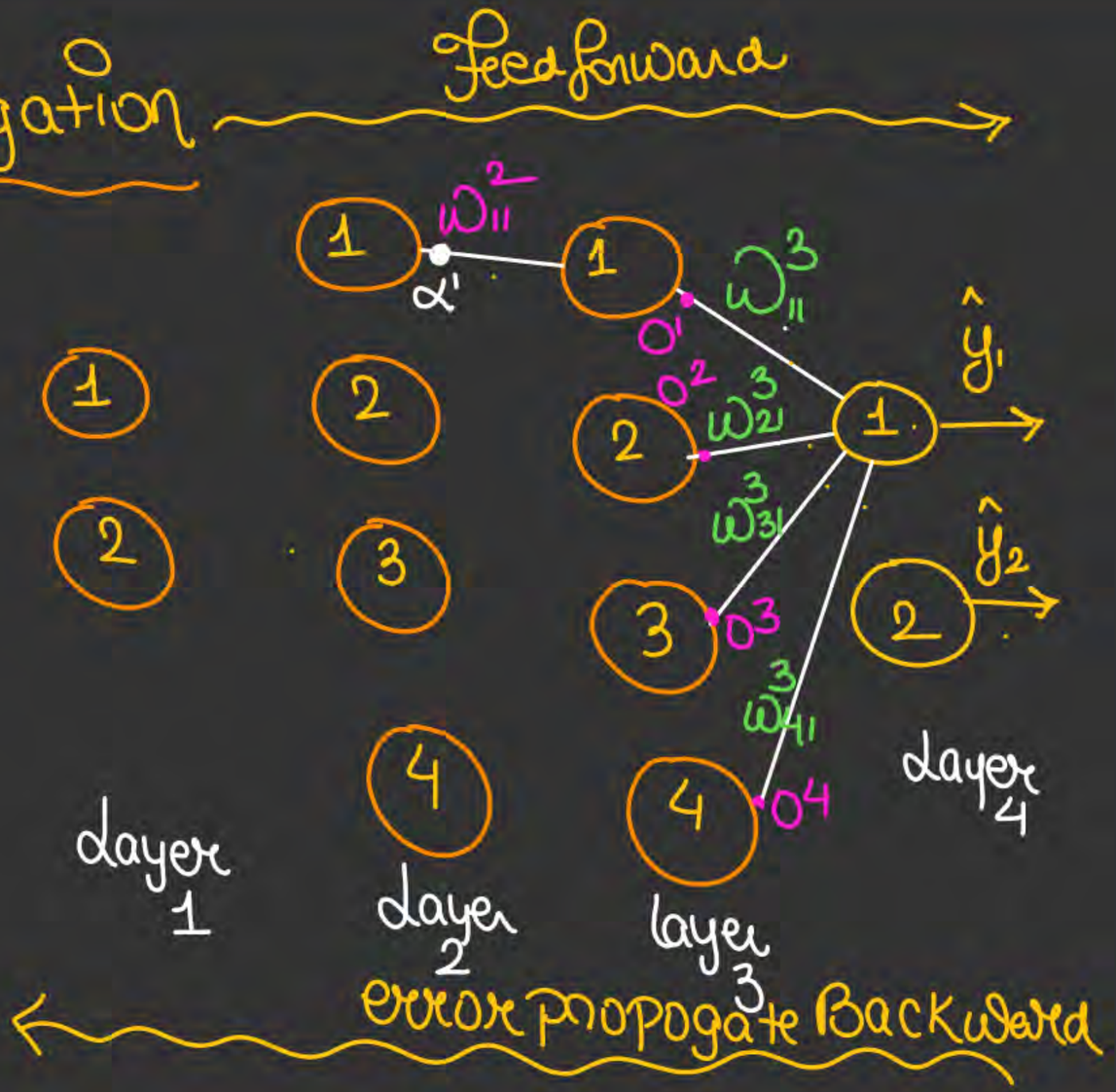
$$\frac{\partial L}{\partial w_{21}^3} \Rightarrow \left[-(y_1-\hat{y}_1)\,\sigma'(\quad)\,O^2\right]$$



Why it is called back propagation

$$\mathcal{L} = \frac{1}{2}\begin{bmatrix}(y_1-\hat{y}_1)^2 \\ (y_2-\hat{y}_2)^2\end{bmatrix}$$

$$\frac{\partial L}{\partial w_{11}^2} \Rightarrow \underbrace{\frac{\partial L}{\partial \hat{y}_1}}_{-(y_1-\hat{y}_1)} \cdot \underbrace{\frac{\partial \hat{y}_1}{\partial w_{11}^2}}_{\frac{\partial \hat{y}_1}{\partial o_1}\cdot\frac{\partial o_1}{\partial w_{11}^2}} + \underbrace{\frac{\partial L}{\partial \hat{y}_2}}_{-(y_2-\hat{y}_2)} \cdot \frac{\partial \hat{y}_2}{\partial w_{11}^2}$$



Perceptron learning

Practise

Question 2: You are training a multi-layer perceptron (MLP) with 3 hidden layers. The first hidden layer has 50 neurons, the second has 30, and the third has 20. How many total neurons are in the hidden layers?

A. 50
B. 80
C. 100
D. 150

Perceptron learning

Practise

Question 3: In a feed-forward neural network, you have an input layer with 15 features. The first hidden layer has 10 neurons, and the second hidden layer has 5 neurons. How many weights are there in the connections between the input and the first hidden layer?

A. 150
B. 155
C. 165
D. 170

Practise

Question 5: In a 4-fold cross-validation, what percentage of the data is used for testing in each fold?

A. 25%
B. 30%
C. 40%
D. 50%

Perceptron learning

Practise

Question 12: You have a multi-layer perceptron (MLP) with an input layer of 40 features and a hidden layer with 30 neurons. How many weights are there connecting the input to the hidden layer, not considering bias terms?

A. 1200
B. 30
C. 40
D. 70

Perceptron learning

Practise

The primary difference between FFNN and recurrent neural networks (RNN) is:

- A) FFNNs have memory while RNNs do not.
- B) RNNs have memory while FFNNs do not.
- C) FFNNs are used for time-series data.
- D) RNNs can only have one hidden layer.

Perceptron learning

Practise

In a feedforward neural network, the information moves:

- A) Forward and backward.
- B) Only forward, from input to output.
- C) In a cyclic manner.
- D) In any random direction.

Perceptron learning

Practise

In a feedforward neural network, the information moves:

- A) Forward and backward.
- B) Only forward, from input to output.
- C) In a cyclic manner.
- D) In any random direction.

Perceptron learning

Practise

A feedforward neural network:

- A) Allows connections to form cycles.
- B) Does not allow connections to form cycles.
- C) Is the same as a recurrent neural network.
- D) Only consists of one hidden layer.

Perceptron learning

Practise

The universal approximation theorem states that:

- A) A single-layer perceptron can approximate any function.
- B) An MLP with at least one hidden layer can approximate any continuous function given enough neurons.
- C) An MLP with two hidden layers can solve the XOR problem.
- D) MLPs are only useful for classification tasks.

Perceptron learning

Practise

In a multi-layer perceptron, the activation function used in hidden layers is typically:

- A) Linear
- B) Step
- C) Non-linear (e.g., ReLU, Sigmoid, Tanh)
- D) None of the above

Perceptron learning

Practise

Which of the following is true about single-layer perceptrons?

- A) They can solve XOR problems.
- B) They can have multiple hidden layers.
- C) They are the building blocks of deep neural networks.
- D) They can only solve linearly separable problems.

Perceptron learning

Practise

The learning rule used by the single-layer perceptron is known as:

- A) Hebbian Learning
- B) Gradient Descent
- C) Delta Rule
- D) Backpropagation

advantage

- Can work on big data.
- Can learn Complex pattern easily
- Regression & Classification

disadvantage

- many hyperparameters are to be tuned
  - No of layers/ No of neurons.
- need large data set
- large No of Computation
- Over fitting

Advantage of NN

Advantages of Neural Networks

1.Ability to Learn Non-Linear Relationships:

1. Neural networks can model complex, non-linear relationships between inputs and outputs, which traditional linear models might miss.

2.Adaptability:

1. Neural networks can adapt to changing data and improve performance with more data, making them suitable for dynamic environments.

3.Parallel Processing:

1. Due to their architecture, neural networks can perform multiple computations simultaneously, enabling faster processing, especially with modern hardware like GPUs.

Advantage of NN

1. **Versatility:**
   1. Neural networks can be applied to a wide range of problems, including classification, regression, clustering, and more.
2. **Handling High-Dimensional Data:**
   1. Neural networks are effective at handling large volumes of data with many features, which is common in modern datasets.

Disadvantages of NN

Disadvantages of Neural Networks

1. **Computational Cost:**
   1. Training neural networks, especially deep neural networks, requires significant computational resources and time.
2. **Data Requirements:**
   1. Neural networks typically require large amounts of data to perform well, which can be a limitation when data is scarce.
3. **Overfitting:**
   1. Neural networks can easily overfit to the training data, especially if they are too complex or if there is insufficient data.
4. **Lack of Interpretability:**
   1. Neural networks are often considered "black boxes" because their internal workings are not easily interpretable, making it difficult to understand how they make decisions.

Perceptron learning

Disadvantages of NN

1.Hyperparameter Tuning:

1. Neural networks require careful tuning of hyperparameters (e.g., learning rate, number of layers, number of neurons per layer), which can be complex and time-consuming.

2.Requirement of Specialized Hardware:

1. Effective training and deployment of neural networks often require specialized hardware, such as GPUs or TPUs, which can be expensive.

3.Difficulty in Training:

more hidden layers

1. Training deep neural networks can be challenging due to issues like vanishing and exploding gradients, which require sophisticated techniques to mitigate.

$\Rightarrow \frac{\partial L}{\partial w} \approx 0$

Perceptron learning

What is cross entropy loss function

1.The cross-entropy loss function, also known as log loss, is a commonly used loss function in classification problems, particularly in binary and multiclass classification tasks involving neural networks. It measures the performance of a classification model whose output is a probability value between 0 and 1.

$$L = -\left[y\log(p) + (1-y)\log(1-p)\right]$$

where:

- y is the true label (0 or 1).
- p is the predicted probability of the instance being in class 1.

THANK - YOU